

Heterogeneous Consequences of Educational Mismatch: Evidence from Indonesia

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Abstract

This paper examines the heterogeneous wage effects of educational mismatch across formal and informal employment in Indonesia. Unlike developed economies, informal employment dominates the labor market. This institutional characteristics may shape the wage consequences of educational mismatch through differences in bargaining power and credential requirements. Using the Propensity Score Matching (PSM), this paper sheds light on several findings. First, overeducated workers earn approximately 11 percent lower wages relative to adequately matched workers, while undereducated workers receive no wage premiums. Second, substantial heterogeneity exists across gender and formality status. Male undereducated workers in the formal sector obtain wage premiums, whereas female workers do not. In addition, overeducated female workers in the informal sector experience larger wage penalties than male workers.

Keywords: Wage Premium, Wage Penalty, Educational Mismatch, Gender, Formal Sectors.

JEL Classifications: I24, J16, J31, C52.

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1 Introduction

Educational mismatch has been one of the crucial challenges in the labor market. The prevalence of educational mismatch raises crucial concerns as it may reflect inefficient allocation of human capital and lower returns to educational investment (Leuven and Oosterbeek, 2011). The overeducated workers, for instance, raises welfare costs through lower wages relative to adequately matched workers (i.e., wage penalties). In contrast, the undereducated workers may gain wage premiums but such mismatches may also indicate inefficiencies in worker-job allocation.

The outcomes of educational mismatch heavily depend on labor market institutions. Sattinger and Hartog (2013), for instance, emphasize the role of bargaining power in shaping wage premium and penalties associated with educational mismatch. Sun (2026) also highlights differences in labor market rigidity as a crucial determinant of wage penalties and premiums heterogeneity among OECD countries. Despite the importance of institutions, however, evidence on the role of job formality status on premiums and penalties remains limited. This distinction is particularly relevant because formal and informal sectors differ substantially in terms of bargaining power, and credential requirements.

This paper attempts to replicate the work by Sun (2026), and extends and applies the concept to Indonesian case study by incorporating with heterogeneous wage effects of educational mismatch across formal and informal employment in Indonesia. In this regard, Indonesia provides an interesting setting for the analysis because of distinct labor market structure from most of the developed economies. Based on 2025 Indonesian Labor Force Survey, 59 percent of workers in Indonesia are informal. In addition, unlike developed economies, undereducated workers outnumber overeducated workers in Indonesia (Allen, 2016). Undereducated workers account for 32 percent of total workers, while only 18 percent are overeducated.

The dominance of informal employment may shape the wage implications of educational mismatch. Compared with formal workers, informal workers are less likely to be covered by formal labor contracts, collective bargaining arrangements through union, and minimum wage protection (International Labour Organization, 2023). The role of educational credentials and wage determination may therefore relatively weak for informal workers. Consequently, the wage penalties and premiums in Indonesia may have different pattern from those usually observed in developed economies.

This paper address two main research questions: what are the impacts of educational mismatch on labor market outcomes in Indonesia and how these effects may differ across formal and informal sectors? Estimating causal effects of educational mismatch, however, requires a careful identification strategy since mismatch status relies on both observable and unobservable characteristics. To overcome this issue, this paper employs Propensity Score Matching (PSM) with Nearest Neighbor Matching (NNM), similar with empirical strategy used by Sun (2026). PSM can address potential selection problem by constructing comparable treatment and control groups with comparable observable characteristics.

This paper sheds light on several crucial findings on the applications to Indonesian case study. First, I find that overeducated workers earn 13.5 percent less than their adequately matched counterpart, while undereducated workers receive no wage premiums. The absence of wage premiums for undereducated workers may reflect low bargaining power dictated by lack of outside options and weak union. However, male undereducated workers in formal sector receive substantial wage premiums while female undereducated workers do not. This suggests

asymmetric bargaining power between male and female workers. Lastly, this paper documents that overeducated female workers in informal sectors experience larger penalties than that of male workers. In contrast, overeducated female workers in formal sectors experience no penalties while male workers are penalized.

This paper contributes to the literature in several ways. First, to the best of my knowledge, empirical works addressing similar questions are still limited. For instance, Aleksynska and Kolev (2021) examined the linkage between educational mismatch and informality from 15 countries in Latin America and Africa. However, this paper did not investigate its impact on wages. For the case of Indonesia, Wicaksono et al. (2023) examined the educational mismatch and wage penalties but did not account for informality aspects. Moreover, Paramitasari et al. (2024) merely investigated how agglomeration affect the prevalence of educational mismatch in Indonesia. Second, this paper provides evidence on how the wage consequences of educational mismatch differ between developing and developed economies by emphasizing the role of labor market formality. In particular, this paper highlights how formal and informal employment may shape the wage penalties and premiums associated with educational mismatch.

This paper is organised as the following sections. Section 2 reviews the literature related to educational mismatch consequences. Section 3 discusses the empirical strategy I use in this paper including data, variables, econometric specifications, and robustness check strategy. Lastly, Section 4 and 5 are results and conclusion, respectively.

2 Literature Review

This paper is closely related to research done by Sun (2026). More specifically, his paper raises fundamental questions: *What are the consequences of educational mismatch on labor market outcomes and how the effects differ by gender and may vary across countries?* To that end, the PIAAC data for the first two survey rounds (2011-2014) is exploited. This large scale international assessment dataset administered by the OECD covers wide range of data from educational attainment and cognitive skills to employment and job characteristics. Consistent with the objective, this research specifically restricts its observations on employed workers aged 16-65 who have completed their education and established in the labor markets, excluding those who still in school-work transition.

Methodologically speaking, there several main challenges to capture the causal effects of educational mismatch on labor market outcomes. The first challenge is related to how it defines educational mismatch. Many of previous works relied on self-assessment which was convenient yet bias. In this paper, educational mismatch is defined by identifying credentials gap and the discrepancy of individuals' skill numeracy scores – individuals are overeducated (undereducated) if their credentials exceed (fall below) job requirements and their numeracy scores lower (higher) than their occupation's median. This indicates both overeducated and underutilized skills. Second, endogeneity problems. This could arise because from workers' selection bias into educational mismatch due to unobservable heterogeneity. To overcome this issue, Sun (2026) utilizes Heckman two step correction for sample selection bias and Propensity Score Matching (PSM).

The paper sheds light on several crucial findings. First, overeducated workers receive 13.8 percent of wage penalties while undereducated workers obtain about 10 percent of wage premiums. Second, the adverse effects of overeducated are also larger for women compared to

men. More specifically, women perceive 14.3 percent wage penalties for being overeducated workers, while only 12.6 percent for men. On the other hand, men obtain 11.3 percent wage premiums which is larger than 7.5 percent wage premiums received by women. Third, both wage premiums and penalties vary significantly across OECD countries. Wage penalties for overeducated workers vary from 30.7 percent in Britain to 6.9 percent in Slovakia. On the other hand, wage premiums vary from 24.1 percent in Chile to 3.7 percent in Israel.

Apart from seminal work by (Sun, 2026), the literature on educational mismatch is one of well-established fields in labor economics. This is primarily because educational mismatch has become one of critical problems in many countries. Duncan and Hoffman (1981) were among the first who observed educational mismatch. Using the US Panel Survey of Income Dynamics (PSID) in 1976, they found that more than 40% of workers attained education level higher than the job requirements (i.e., overeducated) and 12% of workers have education lower than the job requires (i.e., undereducated). Leuven and Oosterbeek (2011) estimated that the average of overeducated and undereducated workers in Latin America accounted for 24 percent and 21 percent, respectively. In China, Ren and Miller (2012) found that incidence of overeducation amounted to 27 percent.

Despite that educational mismatch has been one of the most discussed issues in labor economics, however, studies related to the case of Indonesia remain limited. Wicaksono et al. (2023), for instance, examined the educational mismatch and wage penalties but did not account for informality aspects. Moreover, Paramitasari et al. (2024) merely investigated how agglomeration affect the prevalence of educational mismatch in Indonesia but did not address implications on wage. In addition, none has formally addressed the role of formality in shaping the wage implications associated with educational mismatch. Aleksynska and Kolev (2021) examined the linkage between educational mismatch and informality from 15 countries in Latin America and Africa. However, this paper did not investigate its impact on wages.

3 Empirical Strategy

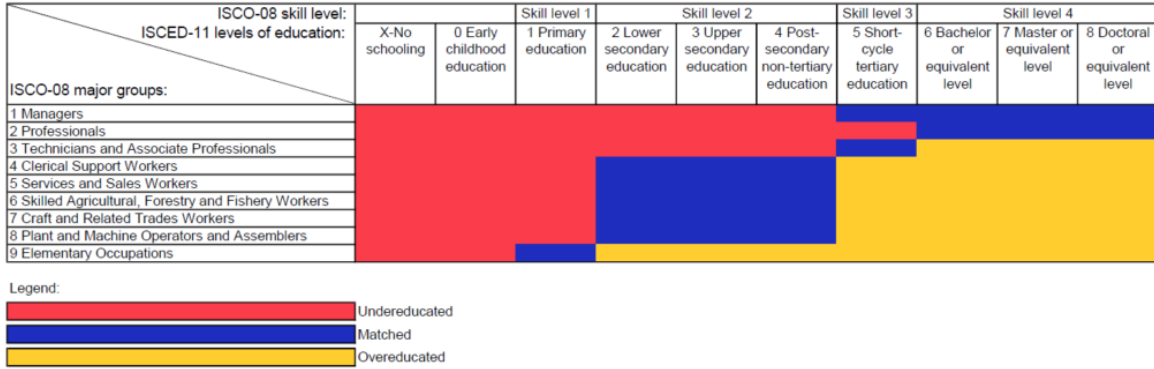
3.1 Data and Variables

I exploit survey data from the Indonesia’s National Labor Force Survey in 2024. This data surveys more than 750,000 samples and covers a wide range of variables required to address my research questions including education attainment, wages, informality, and vocational training experiences.¹

As in Sun (2026), our primary challenges are to define educational mismatch and to control selection bias. For the mismatch measurement, I use a well-established definition from International Labor Organization (ILO). More specifically, ILO specifically categorizes status of educational mismatch according to ISO-08 occupational categories and ISCED-11 levels of education. This approach is commonly used for the case of Indonesia, as in Wicaksono et al. (2023), Paramitasari et al. (2024), and Allen (2016). Figure (1) displays the definition of educational mismatch by ILO. The occupational categories consist of eight major groups, while education level corresponds to three skill levels.

¹Many related works also utilized this data set such as Wicaksono et al. (2023); Allen (2016); Paramitasari et al. (2024).

Figure 1: Mismatch by Level of Education



Source: International Labor Organization (ILO)

Notes: Blue box represents matched education level, red denotes the undereducated, and yellow box is for overeducated.

The approach I use is different from Sun (2026). Specifically, he uses self-assessment (SA) and Realised Match (RM) approaches. The former is obtained from the questionnaire in PIAAC survey which specifically asks respondents to report educational requirements required for their jobs or to assess their match quality. This approach has its own advantages and drawbacks. One of the advantages is that it results in more stable estimates (Leuven and Oosterbeek, 2011). However, bias due to subjectivity may become a problem. In contrast, the RM approach define mismatch status statistically using the central tendency (mean or mode). This approach provides more objective benchmark (Kang and Hong, 2025). Due to data limitations, I will use RM approach, instead of SA, for robustness checks.

Table 1: Disaggregated Educational Match Status

	Matched	Overeducated	Undereducated	Observations
<i>All Samples</i>	47.85%	14.59%	37.55%	261,305
<i>Gender</i>				
Female	47.20%	12.24%	40.56%	96,149
Male	48.23%	15.96%	35.81%	165,156
<i>Formality</i>				
Informal	54.40%	22.65%	22.95%	91,311
Formal	44.13%	10.01%	45.86%	49,233

Source: Indonesia National Labor Force Survey in 2023. See Table (A.5) for summary statistics of control variables.

Table (1) presents the distribution of educational mismatch in Indonesia by formality status, gender, and sector. Overall, approximately 14.6 percent of workers are overeducated, 37.6 percent are undereducated, and 47.9 percent are adequately matched. Different from the pattern commonly found in developed countries, undereducation is substantially more salient than overeducation in Indonesia. By formality status, formal workers exhibit relatively lower

mismatch than informal sectors. Specifically, the incidence of undereducation outnumbers the overeducation for informal workers, while for formal workers, the incidence of under and overeducation is relatively similar. By gender, male workers are more likely to be overeducated, whereas female workers exhibit considerably higher rates of undereducation.

3.2 Econometric Specification

To estimate the causal effects of educational mismatch, there are at least two challenges we need to anticipate. First, potential selection bias problem. The process in which workers end up in over- or under-educated occupations is not random. They are mostly determined by both observable and unobservable characteristics (Bauer, 2002; Sun, 2026). Second, the unbalanced number of observations between treated groups (overeducated and undereducated) and controlled group (educationally matched) could potentially lead to reduced statistical power and unstable estimates of OLS.

Sun (2026) utilises Heckman estimation with two-step correction for the baseline estimates. However, it may remain bias because it controls for employment selection not the selection bias on educational mismatch status. These two biases are conceptually different. The former controls the selection bias between being unemployed and employed, while the later is related to the choice of working at the educationally matched occupations or not.

To overcome these problems, this paper uses Propensity Score Matching (PSM). Specifically, PSM can address potential selection problem by constructing comparable treatment and control groups with comparable observable characteristics (Rosenbaum and Rubin, 1983). PSM also could address the problem related unbalanced samples of treated and controlled groups. Specifically, PSM algorithm can perform one-to-many matching to pair multiple control with a single treated observation (Valojerdi and Janani, 2018). For the baseline estimation I use PSM with Nearest Neighbor Matching (NNM).

Estimating the Average Treatment Effect of the Treated (ATET) using PSM consists of two stages. The first stage is to estimate Propensity Score using binary regression.

$$\Pr(D_i = 1|\mathbf{X}_i) = F(h(\mathbf{X}_i)) \quad (1)$$

where D_i denotes educational mismatch status, $F(\cdot)$ is (Logit) cumulative probability distribution, and $\mathbf{X}_i'$ is vector of control variables. Control variables are Years of Schooling $_i$, Age $_i$, Age $_i^2$, Experience $_i$, Experience $_i^2$, Informality $_i$, Training $_i$, Apprenticeship $_i$, Dummies of Sectors (i.e., Agriculture, Manufacturing, Services), and Dummies of Provinces. See Table A.5 for details of summary statistics.

PSM critically relies on assumption requiring the selection operates through the observables. Therefore, comprehensive control variables in the first stage estimation becomes crucial. In this paper, I use a set of control variables aiming to minimising the residual selection bias which includes employment and geographic characteristics, training or apprentices background, and demographic factors.

The second stage is to calculate ATET which is defined as follows:

$$\text{ATET} = \frac{1}{N^T} \sum_{i \in T} \left(Y_i^T - \sum_{j \in C(i)} w_{ij} Y_j^C \right) \quad (2)$$

where N^T is number of treated observations, $C(i)$ is set of controls for treated i , $w_{ij} = 1/M$ represents weight for nearest neighbor M . For the main estimation, I use three nearest neighbors.

3.3 Robustness Strategy

This paper exploits two robustness check strategies. First, I set nearest neighbor matching equals to five instead of three. This checks whether the main findings are sensitive to the choice of matching algorithm parameters. Second, I use Realised Matching (RM) as alternative measurement for educational mismatch. Specifically, I exploit statistical mode to define the educationally matched occupations. If the education level of a worker is higher (lower) than the matched level, the overeducation (undereducation). For instance, if the statistical mode for managers is upper secondary education, then workers with lower (higher) educational level are categorised as overeducated (undereducated).

Table 2: Disaggregated Educational Match Status (Realised Matching)

	Matched	Overeducated	Undereducated	Total
<i>All Samples</i>	40.43%	24.67%	34.90%	261,305
<i>Gender</i>				
Female	39.57%	19.43%	41.00%	96,149
Male	40.94%	27.71%	31.35%	165,156
<i>Formality</i>				
Formal	48.56%	26.95%	24.49%	94,746
Informal	35.81%	23.37%	40.82%	166,559

Source: Indonesia National Labor Force Survey in 2023.

Table (2) summarises the distribution of educational mismatch status according to gender and formality status from RM approach. Overall, around 40.5 percent of workers are educationally matched, which is 7 percentage points lower than estimate calculated from ILO-based approach. In contrast, the proportion of overeducated workers is about 25 percent, or nearly 10 percentage points higher than the ILO-based approach. The discrepancy between RM and ILO-based approach may arise because the educational requirements derived from statistical mode within occupations tend to be lower than those implied by ILO's definition. Such a difference may reflect Indonesian-specific labour market characteristics that are not fully captured by ILO's definition.

4 Results

This section discusses two main results. First, I elaborate on the replication results from Sun (2026). For this section, I use the same OECD dataset and estimate the wage implications of educational mismatch for all OECD samples and disaggregation samples based on gender. Motivated by Sun (2026), the second part of this section discusses the wage implications of educational mismatch in Indonesia, disaggregating the effects according to the gender and formality status.

4.1 Results from Replication (Sun, 2026)

Table (3) reports the replication results from Sun (2026). The augmented Mincer specification with Heckman two step correction for sample selection bias is exploited to estimate the effects of educational mismatch on wage penalties and premiums. The replication results are identical to the original paper. First, I find that inverse Mills ratio λ for all estimations are significant, reflecting non-random selection into employment generates biases. Such a bias can be illustrated by the deviation between OLS and Heckman estimates. As the Table reports, Heckman estimates result in relative moderate effects than that of OLS estimates.

Furthermore, the results show that the inverse Mills ratio for women is larger than both pooled samples and men. This suggests that unobserved characteristics affecting employment selection correlate with wages more strongly for women. This is likely because women face stricter labour force participation selectivity due to family responsibilities, reservation wages, or occupational references.

Table 3: Heckman Estimation Results (OECD Countries)

	(1)	(2)	(3)	(4)	(5)	(6)
Overeducation	-0.140*** (0.01)	-0.138*** (0.01)	-0.130*** (0.01)	-0.126*** (0.01)	-0.145*** (0.01)	-0.143*** (0.01)
Undereducation	0.105*** (0.01)	0.099*** (0.01)	0.117*** (0.01)	0.113*** (0.01)	0.092*** (0.01)	0.075*** (0.01)
<i>Inverse Mills Ratio</i> λ		0.291** (0.14)		0.342* (0.18)		0.605*** (0.22)
Observations	48,174	51,972	23,799	25,782	24,375	26,190
Estimator	OLS	Heckman	OLS	Heckman	OLS	Heckman
Samples	All	All	Men	Men	Women	Women
Control Variables	All	All	All	All	All	All
Country Dummies	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Asterisks represents probability value of significance * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Numbers in parentheses are standard errors. This table is a replication of Table 3 in Sun (2026). I add OLS regressions for coefficient comparison with Heckman Selection.

From the baseline Heckmen estimates, there are at least two findings that can be highlighted. First, overeducated workers receive 13.8 percent of wage penalties while undereducated workers obtain about 10 percent of wage premiums. Second, the adverse effects of overeducated are also larger for women compared to men. More specifically, women perceive 14.3 percent wage penalties for being overeducated workers, while only 12.6 percent for men. On the other hand, men obtain 11.3 percent wage premiums which is larger than 7.5 percent wage premiums received by women.

Despite these findings, however, Heckman estimation with two-step correction may remain bias because it controls for employment selection not the selection bias on educational mismatch status. These two biases are conceptually different. The former controls the selection bias between being unemployed and employed, while the later is related to the choice

of working at the educationally matched occupations or not.² This motivates this paper to utilise Propensity Score Matching (PSM) instead of Heckman for the extensions.

4.2 Results from Extensions

4.2.1 Baseline Results

This section elaborates on the wage implications of educational mismatch in Indonesia. Figure 2 and 3 respectively report the wage implications for overeducated and undereducated workers. For all samples, I find that overeducated workers earn 13.5 percent less than their adequately matched counterpart. Interestingly, I find that overeducated workers in formal sectors are equally punished of those who work in informal sectors. Specifically, overeducated workers in formal and informal sectors earn about 13 percent less than the matched workers.

However, I find gender-based heterogeneity outcomes. In the absence of discriminatory factors, outcomes between male and female workers should be indifferent. In contrast, I find the outcomes are different according to gender status. The result shows that overeducated male workers experience relatively large wage penalties while female workers experience lower wage penalties. This indicates that educational credentials may play a more prominent role in wage determination for male than that for female. This finding implies that cost of working in lower-level job for female is low which may weaken incentive to transition into educationally matched occupations, and, consequently, intensify misallocation problem in the labor market.

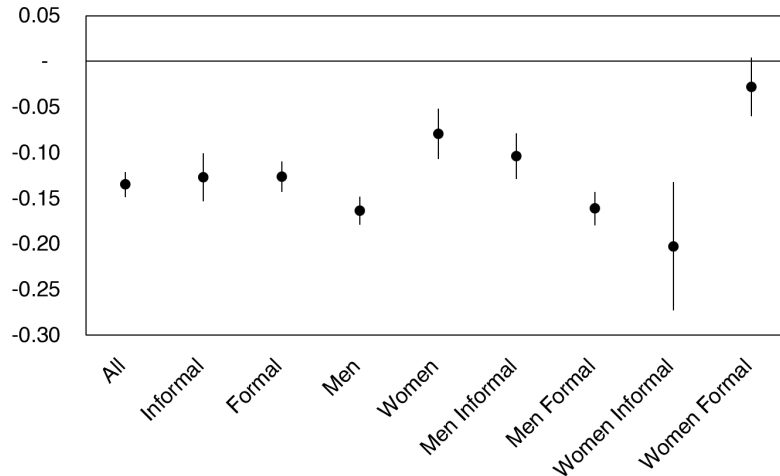


Figure 2: Wage Effects of Overeducation

Notes: This figure displays the coefficients of overeducation on log of wages. The whisker represents the 5% significance level. Positive value indicates wage premium while negative value is for wage penalties. I use PSM-NNM with $n = 3$ to estimate ATET of educational mismatch. Mincerian specification with control dummy for provinces and sectors is used to estimate propensity score.

Furthermore, I also conduct more detailed disaggregation analysis combining formality and gender status. For male workers, I find that overeducated male workers in formal sectors

²See Appendix A.2 for additional estimates from PSM.

get larger wage penalties than those who in informal sectors. This difference can simply be explained by the distinct institutional features between formal and informal sectors. However, I find a rather unusual finding on the overeducated female workers. Specifically, overeducated female workers in informal sectors have larger wage penalties than those who in formal sectors. This finding may reflect low bargaining power of overeducated female workers in informal sectors dictated by lack of outside options.

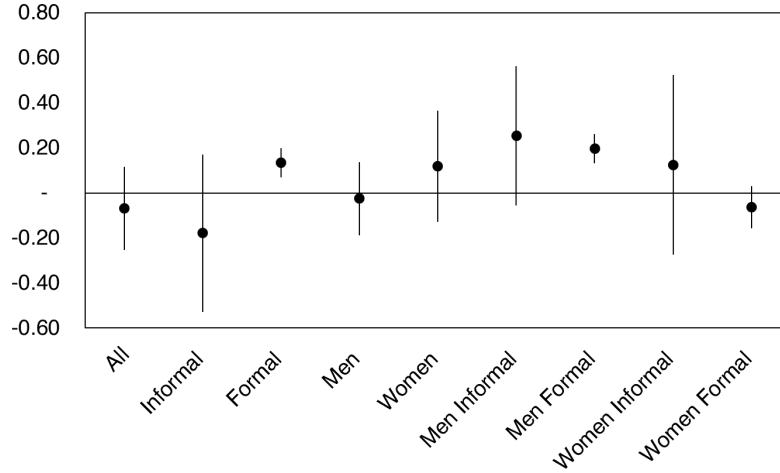


Figure 3: Wage Effects of Undereducation

Notes: This figure displays the coefficients of undereducation on log of wages. The whisker represents the 5% significance level. Positive value indicates wage premium while negative value is for wage penalties. I use PSM-NNM with $n = 3$ to estimate ATET of educational mismatch. Mincerian specification with control dummy for provinces and sectors is used to estimate propensity score.

Furthermore, I find that undereducated workers in Indonesia gain no wage premiums. This results contrast the common findings in developed countries. Such a striking difference, I argue, due to the dominance of informality in Indonesian labour market. As Figure (3) reports, undereducated workers in informal sector receive no wage premiums while those who work in formal sector obtain around 13.4 percent wage premiums. In addition to this finding, I also find that female workers receive no wage premiums, while male undereducated workers in formal sector receive almost 20 percent wage premiums. This finding highlights two important indications of Indonesian labour market. First, absence of wage premiums may reflect weak wage bargaining power for informal workers (Sattinger and Hartog, 2013). This characterises informal sector in which union does not exist. However, second, different outcomes between male and female workers in formal sectors imply male-favouring bargaining in formal sectors.

4.2.2 Robustness Checks

To check the consistency of our main findings, this paper conducts two robustness checks by using alternative definition of educational mismatch and using different number of nearest neighbour matching $n = 5$. Table (4) reports the results from our robustness checks.

First, the main findings are broadly robust to the choice of the number of nearest neighbours. Increasing the matching parameter from $n = 3$ to $n = 5$ does not substantially alter the estimated coefficients. For instance, the wage penalties for overeducated workers based on the ILO definition change only slightly from -0.135 to -0.127 . Similarly, the realised matching estimates remain relatively stable, suggesting that the baseline results are not sensitive to the matching specification. Second, the realised matching (RM) approach produces larger wage penalties for overeducated workers compared to the ILO-based definition. For all samples, the overeducation penalty increases from around 13 percent under the ILO definition to more than 33 percent under RM. This indicates that mismatch measured using occupation-specific educational norms may capture stronger forms of skill underutilisation.

Table 4: Robustness Checks

	Overeducation				Undereducation			
	ILO Definition		Realised Matching		ILO Definition		Realised Matching	
	$n = 3$	$n = 5$	$n = 3$	$n = 5$	$n = 3$	$n = 5$	$n = 3$	$n = 5$
All	-0.135 (0.007)	-0.127 (0.007)	-0.333 (0.011)	-0.328 (0.011)	-0.069 (0.094)	-0.061 (0.081)	-0.095 (0.156)	-0.139 (0.123)
Informal	-0.127 (0.013)	-0.113 (0.013)	-0.376 (0.034)	-0.374 (0.032)	-0.179 (0.178)	0.062 (0.140)	-0.023 (0.170)	-0.108 (0.135)
Formal	-0.126 (0.009)	-0.118 (0.008)	-0.070 (0.009)	-0.071 (0.008)	0.134 (0.033)	0.130 (0.031)	-0.033 (0.069)	-0.055 (0.054)
Men	-0.164 (0.008)	-0.153 (0.007)	-0.333 (0.014)	-0.328 (0.013)	-0.026 (0.083)	-0.018 (0.071)	-0.008 (0.096)	-0.049 (0.081)
Women	-0.079 (0.014)	-0.080 (0.014)	-0.271 (0.022)	-0.254 (0.021)	0.118 (0.126)	0.081 (0.111)	0.353 (0.198)	0.462 (0.155)
Men Informal	-0.104 (0.013)	-0.096 (0.012)	-0.379 (0.027)	-0.365 (0.024)	0.253 (0.158)	0.079 (0.124)	-0.044 (0.090)	-0.093 (0.076)
Men Formal	-0.161 (0.009)	-0.150 (0.009)	-0.108 (0.010)	-0.109 (0.010)	0.196 (0.033)	0.193 (0.031)	0.023 (0.058)	0.025 (0.045)
Women Informal	-0.203 (0.036)	-0.183 (0.034)	-0.500 (0.114)	-0.469 (0.106)	0.124 (0.203)	0.248 (0.159)	0.149 (0.231)	0.099 (0.186)
Women Formal	-0.028 (0.016)	-0.025 (0.015)	0.023 (0.018)	0.023 (0.016)	-0.064 (0.047)	-0.066 (0.044)	-0.050 (0.110)	-0.024 (0.092)

Notes: Standard errors are reported in parentheses. n denotes the number of nearest neighbours used in the matching procedure. The outcome variable is log wage.

Third, substantial heterogeneity across gender and formality status persists throughout all specifications. In particular, overeducated women in informal sectors consistently experience the largest wage penalties, reaching almost 50 percent under RM estimates. In contrast, overeducated women in formal sectors experience negligible or statistically insignificant penalties. This pattern remains stable across both $n = 3$ and $n = 5$. Fourth, the results for undereducation differ sharply by formality status. Undereducated workers in formal sectors consistently receive positive wage premiums, especially for men, while undereducated informal workers

generally receive no significant premiums. This finding is robust across both mismatch definitions and matching specifications, reinforcing the argument that bargaining power and credential recognition are stronger in formal employment. Lastly, although some coefficients vary in magnitude under RM, the pattern remain consistent.

5 Conclusion

By applying the seminal works by Sun (2026), this paper examines the wage implications of educational mismatch in Indonesia. Departing from the fact that informality is prevalent in Indonesia, this paper specifically investigates whether labor market formality shapes the wage penalties and premiums associated with educational mismatch.

Using the dataset from Indonesian National Labour Market Survey in 2023, this paper highlights several crucial findings. First, overeducated workers experience substantial wage penalties relative to adequately matched workers. The baseline estimates show that overeducated workers earn approximately 13.5 percent lower wages. In contrast, undereducated workers generally do not receive wage premiums. This finding differs from the common evidence documented in developed economies and may reflect the dominance of informal employment and relatively weak bargaining institutions in Indonesia.

Second, this paper also finds substantial heterogeneity across gender and formality status. Undereducated workers in formal sectors, particularly male workers, receive positive wage premiums, while informal workers generally do not. Furthermore, overeducated female workers in informal sectors experience the largest wage penalties. These findings suggest that labor market institutions, bargaining power, and credential recognition differ substantially between formal and informal sectors and may shape the wage consequences of educational mismatch. Lastly, these findings from the baseline estimates are consistent across alternative matching specifications and alternative mismatch definitions.

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A Appendices

A.1 Summary Statistics

Table A.5: Summary Statistics

Variable	Obs.	Mean	Std. Dev.	Min	Max
Years of Schooling	261,029	9.007	3.844	3	16
Age	261,305	44.684	13.001	15	98
Work Experience	261,305	28.076	15.923	0	80
Informality Status	261,305	0.637	0.481	0	1
Vocational Training	261,305	0.224	0.417	0	1
Apprenticeship	261,305	0.006	0.076	0	1
Gender	261,305	0.632	0.482	0	1
Agriculture Workers	261,305	0.349	0.477	0	1
Manufacturing Workers	261,305	0.188	0.391	0	1
Services Workers	261,305	0.462	0.499	0	1

Notes: This table reports summary statistics for the main variables used in the analysis. Informality status, vocational training, apprenticeship, gender, and sectoral indicators are binary variables taking value one if the condition is satisfied and zero otherwise.

A.2 Additional Results

Table (A.6) reports the wage implications of educational mismatch using PSM with nearest neighbor matching ($n = 3$) from Sun (2026) dataset. In general, the results show similar patterns with the baseline estimates (in Table 3). Specifically, women experience larger wage penalties of being overeducated than men but receive smaller wage premiums of being undereducated. However, the magnitude of both penalties and premiums are larger (in absolute terms) compared to the baseline estimates using Heckman two-step corrections.

Table A.6: Propensity Score Matching Estimates by Educational Mismatch and Gender

Mismatch Status	Sample	ATT	S.E.	Upper Bound	Lower Bound
Overeducation	All	-0.1933	0.0093	-0.1752	-0.2114
Undereducation	All	0.1654	0.0117	0.1883	0.1425
Overeducation	Male	-0.1817	0.0137	-0.1549	-0.2085
Overeducation	Female	-0.2043	0.0125	-0.1798	-0.2288
Undereducation	Male	0.1763	0.0161	0.2078	0.1447
Undereducation	Female	0.1362	0.0168	0.1692	0.1032

Notes: The table reports the Average Treatment Effect on the Treated (ATT) estimated using Propensity Score Matching with nearest-neighbour matching. Upper and lower bounds correspond to the 95 percent confidence interval.